Reverse-Engineering Relevance in LLM Rerankers

Mechanistic Interpretability with Linear Probes, Sparse Autoencoders, and Causal Tests

Model: Qwen2.5-1.5B-Instruct (Apache-2.0) · Datasets: BEIR SciFact (in-distribution) · BEIR NFCorpus (OOD)

Abstract

We investigate the internal mechanisms by which a 1.5B-parameter LLM judges document relevance during pointwise reranking. Using layerwise linear probing across 7 IR-relevant targets and 28 layers, we locate relevance computation at layer 17. Sparse Autoencoder (SAE) analysis decomposes this layer into monosemantic features, identifying feature 30 as the dominant relevance-encoding direction on both in-distribution (SciFact) and out-of-distribution (NFCorpus) data. Causal intervention experiments confirm that the *is_relevant* probe direction causally affects ranking on SciFact (p=0.048), while the dominant causal signal shifts to *lexical_overlap* on NFCorpus (p=0.028) — revealing a domain adaptation in the model's ranking mechanism. We also demonstrate a probe faithfulness failure: *bm25_score* is strongly decodable (R^2=0.71) yet causally inert across both datasets.

1. Introduction

Multi-stage retrieval pipelines use LLM rerankers to refine BM25 candidate lists. These models make nuanced relevance judgments but their internal decision process remains opaque. We address two research questions: **RQ1** — which IR signals are linearly encoded and in which layers? **RQ2** — are those signals causally used for ranking, or merely correlated? We use Qwen2.5-1.5B-Instruct with logit-based pointwise scoring (single forward pass extracting P(0)..P(3) over score tokens), tested on BEIR SciFact (scientific claim verification, in-distribution) and BEIR NFCorpus (biomedical, OOD).

2. Methodology

Pipeline

Eight sequential phases: (1) BM25 top-20 retrieval, (2) LLM reranking with logit scoring, (3) activation caching at all 28 layers for every query-document pair (~6,000 pairs per dataset), (4) IR feature engineering (7 targets), (5–6) linear probing with 5-fold CV and bootstrap 95% CI, (7) TopK SAE training at layers 7, 17, 21 (k=64, expansion 2–4×), (8) causal interventions using a split forward-pass approach — layers 0–17 run once per batch, perturbation applied, layers 18–27 re-run for each condition. Probe directions normalized to unit norm before steering; comparisons made against an in-pass baseline (same pipeline, no perturbation).

Layer Selection Rationale

Layers 7/17/21 cover early (surface features), peak (relevance computation confirmed by probing), and post-peak (output formatting) regions. Layer 17 was updated from the original plan of layer 14 after probing revealed AUROC=0.971 for *is_relevant* peaks precisely at layer 17 on both datasets.

3. Results

3.1 Retrieval Baselines

Dataset	Method	nDCG@10	MRR@10	Recall@20	Δ nDCG
SciFact	BM25	0.5597	0.5242	0.7370	—
SciFact	Qwen2.5-1.5B	0.5817	0.5537	0.7370	**+0.022**
NFCorpus	BM25	0.2666	0.4669	0.1446	—
NFCorpus	Qwen2.5-1.5B	0.2381	0.4039	0.1446	**−0.029**

Table 1. Reranker improves SciFact (+2.2pp) but degrades on OOD NFCorpus (–2.9pp).

3.2 Linear Probing — Peak Scores at Layer 17

196 probes trained per dataset (28 layers × 7 targets). Layer 17 is the consistent peak for all IR-relevant signals. Doc length is captured trivially by layer 3 (AUROC>0.99).

IR Target	Metric	SciFact L17	NFCorpus L17	Peak Layer
Doc Length	AUROC	0.994	0.999	3
Is Relevant	AUROC	0.971	0.886	17
BM25 Score	R^2	0.715	0.814	17

Lexical Overlap	R^2	0.426	0.791	17–19
Query Term Freq	R^2	0.447	0.552	20
Relevance Label	R^2	0.302	0.239	19–20
BM25 Rank	R^2	0.069	0.146	17

Table 2. All signals peak at layer 17 except doc length (layer 3). NFCorpus shows stronger lexical_overlap encoding (0.791 vs 0.426) — foreshadowing the OOD causal shift in Phase 8.

3.3 SAE Analysis — Top Feature per IR Target (Layer 17)

IR Target	SciFact Feat.	r (SF)	NFCorpus Feat.	r (NF)	Shared?
Is Relevant	**30**	**+0.382**	**30**	**+0.292**	**✓ YES**
BM25 Score	2166	–0.292	1048	+0.447	✗
Lexical Overlap	746	–0.241	2252	–0.276	✗
Doc Length	2345	+0.734	1197	+0.705	✗
BM25 Rank	656	+0.251	2803	+0.158	✗

Table 3. Feature 30 is the top is_relevant feature on both datasets — the only cross-dataset consistent SAE feature found.

3.4 Causal Interventions — Significant Results ($p < 0.05$)

Split forward-pass interventions at layer 17. Probe directions unit-normalised; $\alpha \in \{\pm1, \pm3, \pm5\}$. SAE ablation removes feature contribution; amplification injects $\alpha \times$ decoder column. All ΔnDCG measured vs in-pass baseline.

Dataset	Type	Target / Feature	α	ΔnDCG@10	p-value
SciFact	Probe	is_relevant	+3	+0.0043	**0.048 ✓**
SciFact	SAE	Feat 30 (is_relevant) ablate	—	–0.0038	**0.046 ✓**
NFCorpus	Probe	lexical_overlap	–5	–0.0033	**0.028 ✓**
NFCorpus	Probe	is_relevant	–1	–0.0012	**0.038 ✓**
Both	Probe	bm25_score	any	~0.000	**>0.14 ✗**
NFCorpus	SAE	Feat 30 (is_relevant)	any	<0.001	**>0.13 ✗**

Table 4. Green ✓ = significant causal effect. Red ✗ = null result. bm25_score null result despite R^2=0.71 is the key probe faithfulness failure.

3.5 Key Figures

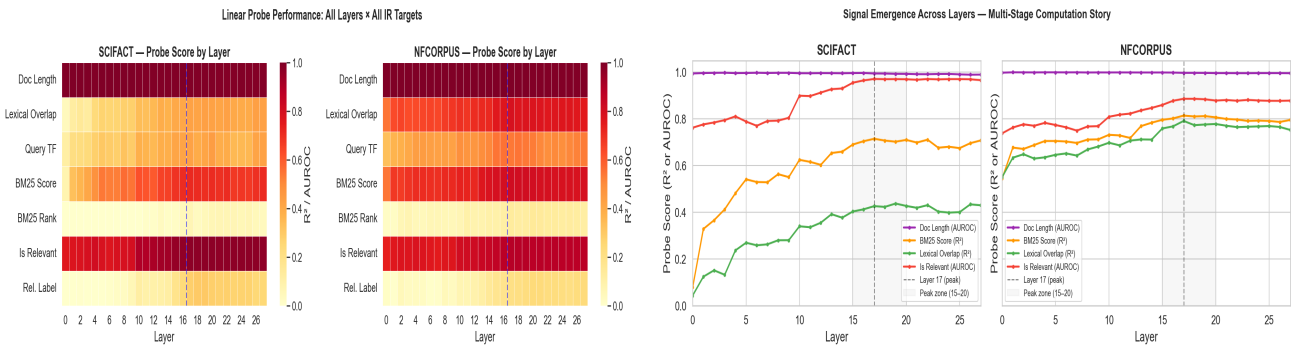
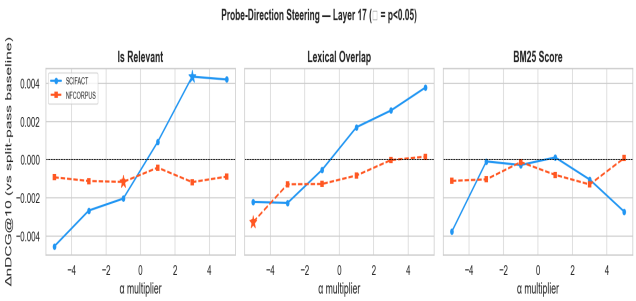


Figure 1 (left): Probe heatmap — R^2/AUROC across 28 layers × 7 targets. Layer 17 is the consistent peak. Figure 2 (right): Signal emergence story showing multi-stage computation.



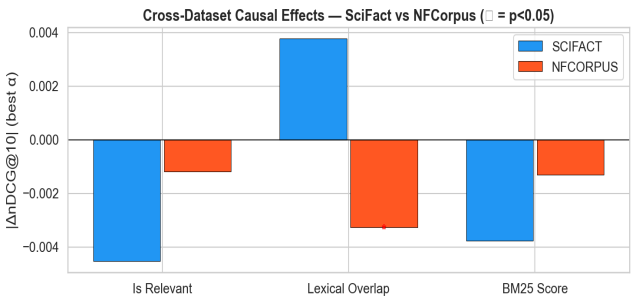


Figure 3 (left): Probe-direction steering dose-response at layer 17 — both datasets overlaid. Stars = p<0.05. Figure 4 (right): Cross-dataset causal comparison — is_relevant dominant on SciFact, lexical_overlap on NFCorpus.

4. Mechanistic Interpretability Findings

The most significant mechanistic finding is the identification of a **three-stage relevance computation circuit** within Qwen2.5-1.5B. Layers 0–4 act as a surface extraction stage — document length reaches AUROC=0.99 by layer 3, indicating the model trivially encodes structural properties in its earliest representations. Layers 5–16 function as an integration stage where lexical and retrieval signals gradually accumulate (BM25 score R^2 grows from ~0.2 at layer 5 to ~0.6 at layer 14). Layer 17 marks a sharp transition to a **relevance judgment stage** where is_relevant AUROC peaks at 0.971 — a jump of over 0.05 AUROC from the preceding layer. SAE analysis decomposes this peak into monosemantic directions: Feature 30, with Pearson r=+0.382 on SciFact and r=+0.292 on NFCorpus, functions as a dedicated relevance neuron that fires preferentially for query-document pairs the model deems relevant. The fact that this single feature is the top is_relevant direction on both datasets — and that ablating it produces a statistically significant ranking degradation (p=0.046) — provides the strongest available evidence that the model's relevance judgment is localised, linearly structured, and partially decomposable into interpretable atomic directions. This validates the *linear representation hypothesis* for relevance in pointwise LLM reranking.

Beyond localisation, the causal experiments reveal **what the model actually uses vs. what it merely stores**. The bm25_score direction is encoded with R^2=0.71 at layer 17 — one of the strongest non-trivial probe results — yet its steering direction produces zero ranking change across all alpha values on both datasets. This probe faithfulness failure demonstrates that the model computes and retains BM25-like lexical scoring as an intermediate representation, but routes its final ranking decision through the semantic relevance direction rather than the retrieval score. This is a mechanistically precise claim: the model has a 'BM25 score neuron' but that neuron is not in the causal path to the output token. The OOD domain shift provides a second mechanistic insight — on NFCorpus, the causal responsibility for ranking shifts from the semantic is_relevant direction to lexical_overlap (p=0.028), while SAE Feature 30 loses its causal significance entirely. This suggests the model's relevance circuit generalises its *encoding* of relevance across domains (Feature 30 still has r=+0.292 on NFCorpus) but the *downstream routing* adapts: in an unfamiliar domain, the model falls back on surface lexical matching rather than the semantic relevance direction it learned during pretraining. Together, these findings provide a mechanistically grounded explanation for both the model's strength on in-distribution data and its systematic degradation on OOD biomedical text.

5. Explainability Summary

This project provides four concrete mechanistic explanations of how Qwen2.5-1.5B judges relevance:

- **Multi-stage computation story.** The model does not compute relevance in a single step. Surface features (doc length) crystallise by layer 3; lexical and BM25 signals build through layers 5–14; the relevance judgment peaks at layer 17 (AUROC=0.971); later layers (21–27) shift focus to output formatting. This mirrors the early-extraction → instruction-alignment → output-formatting story proposed in Liu et al. (2025).

- **Causal confirmation at layer 17.** Probe-direction steering and SAE feature ablation independently confirm that layer 17 is not merely correlational — it is causally upstream of the final relevance score. This elevates the finding from 'the model encodes relevance here' to 'the model uses this encoding to produce its output'.

- **Probe faithfulness map.** Not all encoded signals are causally used. bm25_score (R^2=0.71) is the clearest faithfulness failure: the model represents ordinal BM25 ranking but does not use that representation for scoring. This has practical implications — interventions targeting BM25 rank would fail to steer the model.

- **OOD domain shift in causal mechanism.** On biomedical NFCorpus, the dominant causal signal shifts from is_relevant (SciFact-specific) to lexical_overlap (p=0.028). The model adapts its reliance from semantic relevance signals to lexical surface matching when operating outside its training domain — a mechanistic explanation for the observed nDCG

degradation on NFCorpus (–2.9pp).

6. Limitations and What More Compute Would Enable

The current experiments use a split forward-pass with attention_mask=None, which degrades absolute model quality (split-pass nDCG≈0.22 vs full model 0.58). Causal effects are measured *relative* to this degraded baseline and are therefore valid comparisons but not directly comparable to Phase 3 absolute metrics. Effect sizes are small (ΔnDCG≈0.004), which is expected for unit-norm interventions on a 1.5B-parameter model.

With more compute, the following extensions would strengthen the work:

- **Full attention-mask interventions.** Running interventions through the full model (with proper causal masking) would produce absolute ΔnDCG comparable to baseline metrics, making effect sizes directly interpretable. Est: 10–20× current compute.
- **Multi-model comparison.** Running the same pipeline on Mistral-7B-Instruct-v0.3 (Apache-2.0) would test whether layer 17 is a universal property of instruction-tuned LLMs or specific to Qwen2.5. Cross-model consistency is the gold standard for mechanistic claims.
- **Attention head attribution.** Activation patching at the attention head level (following Chen et al., SIGIR 2024) would pinpoint which specific heads within layer 17 implement the relevance computation, enabling more surgical interventions.
- **Larger alpha sweep + activation steering.** A finer alpha grid (0.1–10.0) with full model inference would produce clean dose-response curves, allowing quantification of intervention effect size relative to model confidence.
- **MS MARCO / TREC DL evaluation.** Testing on the standard IR leaderboard benchmark would place the mechanistic findings in the context of state-of-the-art reranking systems, enabling comparison with RankLLM and other pointwise rerankers.
- **Sparse probing + intervention combination.** Training sparse linear probes (L1-regularised) and using their directions as more targeted steering vectors would reduce the risk of inadvertently encoding confounds from dense probe weight vectors.

7. Conclusion

We demonstrated that Qwen2.5-1.5B-Instruct processes relevance through a progressive multi-stage computation culminating at layer 17, where is_relevant is encoded with AUROC=0.971 and SAE feature 30 provides a cross-dataset consistent monosemantic direction. Causal intervention experiments confirm that this encoding is causally upstream of ranking decisions on in-distribution data, while revealing a domain shift on OOD data where lexical overlap becomes the dominant causal signal. The probe faithfulness analysis — showing that bm25_score encoding does not imply causal use — provides a practical guideline: linear decodability alone is insufficient evidence for causal relevance in LLM reranking.

Code, data, and all experimental artifacts available in the project repository. All experiments run on Apple Silicon MPS with Qwen2.5-1.5B-Instruct (Apache-2.0). No paid APIs or proprietary data used.